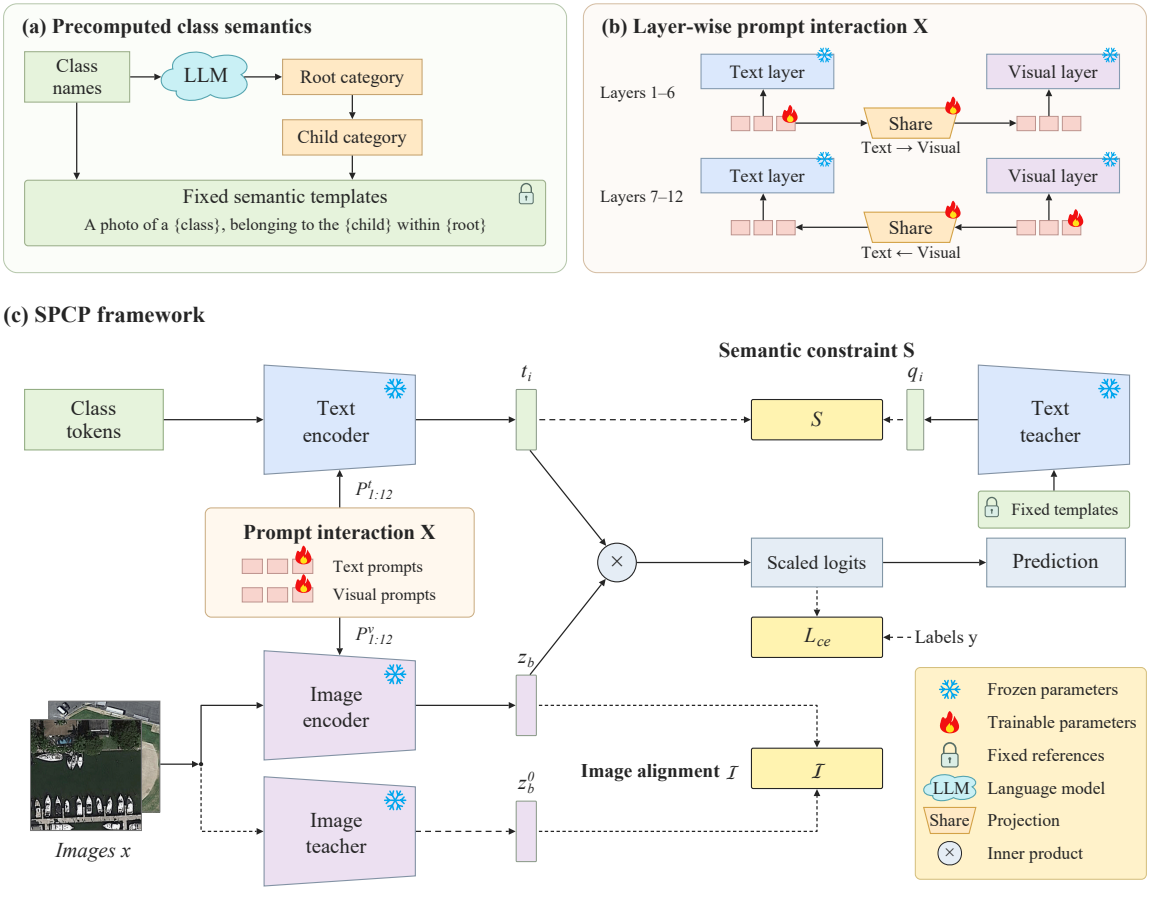

(a) Precomputed class semantics
Class names
LLM
Root category
Child category
Fixed semantic templates
A photo of a {class}, belonging to the {child} within {root}
(b) Layer-wise prompt interaction X
Layers 1–6
Text layer
Visual layer
Share
Text → Visual
Layers 7–12
Text layer
Visual layer
Share
Text ← Visual
(c) SPCP framework
Class tokens
Text encoder
t_i
$P^t_{1:12}$
Prompt interaction X
Text prompts
Visual prompts
$P^v_{1:12}$
Image encoder
z_b
Image teacher
z^0_b
Images x
Semantic constraint S
S
q_i
Text teacher
Fixed templates
Scaled logits
Prediction
L_{ce}
Labels y
Image alignment $\mathcal{I}$
$\mathcal{I}$
Frozen parameters
Trainable parameters
Fixed references
LLM
Language model
Share
Projection
Inner product